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Data Processing Methods for Financing Terrorism: The Role of Microsoft Power BI in Money Laundering Detection

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Abstract

This article explores the imperative landscape of money laundering detection, emphasizing the role of advanced financial monitoring and the integration of Microsoft Power BI for in-depth data analysis. Addressing challenges in fraud detection, the study employs synthetic datasets from Kaggle, overcoming limitations in accessing real financial data. Through a meticulous process, the article demonstrates the power of Power BI in uncovering potential money laundering activities via visual analyses such as stacked column charts, line charts, pie charts, and clustered bar charts. The significance of appropriate data preparation, column selection, and visual representations is highlighted, offering a systematic approach for fraud detection in financial datasets. This research showcases the effectiveness of Power BI as a strategic tool, providing insights and methodologies that contribute to the ongoing discourse on fraud prevention in financial analytics.

Keywords: Money laundering detection; Financing terrorism; Microsoft Power BI; Synthetic datasets; Fraud detection techniques; Visual data analysis; Financial dataset analysis; Data preparation and cleaning.

1. Introduction

Financing terrorism involves providing funds or financial support to terrorist organizations or individuals with the intent to carry out terrorist activities. This support can come from a variety of sources and may involve both

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legitimate and illegitimate financial channels. Governments and international organizations have implemented measures to detect and prevent the financing of terrorism to enhance global security.

Terrorist groups derive funds through illegal activities such as drug trafficking, arms smuggling, and kidnapping. Extortion payments and state sponsorship from certain nations also contribute to their financial resources.

In terms of methods, terrorists employ money laundering techniques to conceal fund origins, and the use of cryptocurrencies adds a layer of anonymity, making it challenging to trace and monitor financial transactions. Regulatory measures include Anti-Money Laundering (AML) laws established by many countries and the implementation of Know Your Customer (KYC) procedures by financial institutions to verify identities and monitor transactions for suspicious activities.

Money laundering poses a substantial financial challenge, involving transactions worth billions of dollars. Detecting these illicit activities proves to be an intricate task, with many automated algorithms frequently generating a notable number of false positives. This often leads to lawful transactions being mistakenly flagged as potential instances of money laundering [1-3]. Conversely, there is a significant apprehension surrounding false negatives, indicating instances where money laundering activities successfully go unnoticed. Criminal entities, as anticipated, deploy substantial efforts to conceal and complicate the identification of their illicit financial activities.

1.1. Money laundering overview

Money laundering represents a significant issue in diverse industries, carrying substantial economic repercussions. The pursuit of fraud detection and prevention is highly valuable. To proficiently conduct research, develop, test, evaluate, measure, and enhance fraud detection techniques, it is crucial to possess in-depth information about the particular domains susceptible to fraud and their distinctive features. Access to publicly accessible data featuring a range of financial transaction scenarios will fulfill these needs, facilitating the comparison and contrast of different approaches to enhance the detection of both fraud and money laundering.

1.2. Investigating the role of Microsoft BI

Regrettably, due to a confluence of factors, including confidentiality imperatives, data integrity concerns, legal stipulations, and adherence to internal protocols, access to pertinent data proves intricate and, in certain instances, unattainable for external researchers within the realm of this scientific inquiry [3]. Even with requisite authorizations, procuring information that encapsulates a diverse array of fraud scenarios—facilitating iterative experimentation with various methodological approaches and the meticulous refinement of parameters within a controlled environment—often presents formidable challenges.

In this research undertaking, synthetic datasets drawn from Kaggle's preeminent web repository will be harnessed. The primary objective of this scientific study is to conduct a comprehensive analysis of synthetic data through the lens of the Microsoft Power BI tool, contributing valuable insights to the scientific discourse on fraud detection.

2. Literature review

2.1. Historical context of money laundering

The historical context of money laundering traces its roots back to ancient times, with instances of individuals disguising the origins of illicitly acquired wealth found in various civilizations. However, the term "money laundering" itself gained prominence in the 20th century.

One of the earliest documented cases of money laundering dates to the prohibition area in the United States during the 1920s and 1930s. Criminal organizations, notably those involved in bootlegging and illegal alcohol trade, sought ways to legitimize their proceeds. This led to the establishment of front businesses and the manipulation of financial transactions to conceal the illicit origins of funds.

2.2. Previous research

The inception of money laundering detection can be traced back to the 1970s when financial institutions began reporting transactions to government authorities (Soltani et al., 2016) [5]. Subsequently, automated banking systems predominantly employed techniques incorporating established rules with specific thresholds.

In the work by Savagea, Wangb, Chouc, and Zhanga (2016) [6], a system is devised that employs network analysis and supervised learning methodologies for the identification of potentially suspicious groups of customers engaged in financial transactions.The study focuses on two specific transaction types: substantial cash deposits and international fund transfers. In recent publications, Martínez-Sa ́nchez and Cruz-García (2020) [7] employ historical data and tree regression techniques to anticipate the risk associated with exposure to money laundering. The dependent variable in their analysis is the customer's status, characterized by the level of risk. The predictors utilized encompass variables such as legal entity, origin, economic activity, seniority, and contracted product.

Meanwhile, Jullum, Løland, and Bang Huseby (2020) [8] leverage. Their predictive model assesses whether a new transaction warrants reporting, considering factors such as sender/receiver information, their past behavior, and transaction history. In their study, Domashova and Mikhailina (2020) found that “For the training of the final model the methods the most significant features in the model are: the type of activity of the organization, the age of the organization, the size of the authorized capital, the composition of the founders, taxes paid, the amount of profit, etc. ” (Discussion section, para 4) [9].

According to projections by Rosha-Salazar et al. (2021) “The shortage of information on historical money laundering and terrorism financing cases means that it is appropriate to treat them as latent variables” (Methodology section, p.3) [10]. Farber (2023) discuss that “The new policy aims to suppress the economic incentive for terrorist activity by freezing financial aid to the PA due to salaries to terrorists. In addition, various criminal and civil provisions were enacted regarding Banks that are active in the PA and manage terrorist accounts” (Introduction section, p.1) [11, 15]. Thommandru and Chakka (2023) [12] emphasize the “AML compliance demands the installation of many procedures and technology solutions that merge KYC data and systems into a single repository ”. Gaviyau et al. (2023) [13-14] highlighted that “Financial penalties are levied on repetitive legal infringements or the failure to effect corrective measures. The study outlines some of the financial penalties issued hereafter” (Discussion section, p.4).

3. Methodology

3.1. Data collection

A notable deficiency exists in the accessibility of financial services datasets, particularly within the nascent domain of monetary transaction data [14]. These datasets hold significant relevance for researchers, specifically those engaged in investigations pertaining to fraud and the detection of money laundering activities.

This dataset proffers a synthetic compilation generated through the utilization of the PaySim simulator as a strategic response to this challenge [14].

Let's write in more detail about the data, what it generally consists of.

Table 1. Description of data.

Column name	Description
Step	Represents a unit of time in the real world, with each step equivalent to one hour. The total number of steps in this simulation is 744, equivalent to a 30-day period.
Type	Denotes the nature of the transaction and includes categories such as CASH-IN, CASH-OUT, DEBIT, PAYMENT, and TRANSFER
Amount	Signifies the transaction amount in the local currency.
NameOrig	Identifies the customer initiating the transaction.
OldbalanceOrg	Reflects the initial balance of the originator's account before the transaction.
OldbalanceOrg	Reflects the initial balance of the originator's account before

	the transaction.
NewbalanceOrig	Indicates the new balance of the originator's account after the transaction.
NameDest	Specifies the recipient customer of the transaction.
OldbalanceDest	Represents the initial balance of the recipient's account before the transaction. Notably, information is absent for customers with names starting with 'M' (Merchants).
NewbalanceDest	Depicts the new balance of the recipient's account after the transaction.
IsFraud	Denotes transactions executed by fraudulent agents within the simulation.
IsFlaggedFraud	Pertains to the business model's objective of monitoring large-scale transfers between accounts and flagging potentially illegal attempts.

In this dataset, an illegal attempt is defined as an effort to transfer an amount exceeding 200,000 in a single transaction.

3.2. Microsoft Power BI integration

One of the primary objectives of Power BI projects was additionally to display and keep track of models on the user interface [15]. As an integral component of Power BI, these reports provide access to a wide range of tools, encompassing interactive data exploration, visualization, Q&A features, integration with diverse data sources, data refresh functionalities, and additional capabilities [16].

Power BI directs these queries to the connector, which acts as a proxy, transmitting the new user's identity to an authorized user through the Azure Active Directory service. It subsequently applies the existing SSAS security permissions based on roles. The connector then examines the local SSAS cube to retrieve the necessary data, optimizing query performance through cached connections [17]. Microsoft Power BI offers business analysts the flexibility to import data from a variety of local sources using either Power BI Desktop or Microsoft Excel. The integration of Microsoft Power BI Personal Gateway facilitates efficient data management and synchronization, ensuring that reports and dashboards in Power BI are consistently kept up to date [18, 23].

Moreover, Power BI seamlessly connects with various cloud services, including Azure SQL Database, Azure SQL Database Auditing, and Azure Stream Analytics. This integration extends the capabilities of Power BI in the Azure environment, enabling the creation of integrated BI solutions without any disruptions. As an example, Azure Stream Analytics can be employed to process streaming data, and the results can be seamlessly exported to Power BI, allowing for real-time updates to dashboards [19-22].

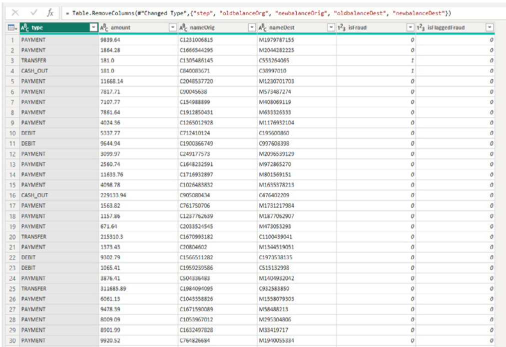
4. Data processing methods

4.1. Exploratory data analysis (EDA)

To get started, we import the data into Microsoft Power BI:

- We initiated the data preparation process by accessing the raw dataset in the Power Query Editor.
- Applying data cleaning methods to remove redundant information and unnecessary columns, focusing on retaining essential features for analysis. The objective is to streamline the dataset and enhance its relevance to the research objectives. Systematically reducing the number of columns to ensure a concise and manageable dataset. The goal is to limit the dataset to a range of 6 to 8 columns, promoting simplicity and facilitating efficient analysis.
- Implementing the appropriate assignment of data types to each column within the dataset. This involves ensuring that numerical values are represented as numeric types, categorical variables are appropriately labeled, and date/time fields adhere to the correct formats.

- Conducting a thorough quality check to validate the integrity of the dataset after the cleaning and data type application steps. This involves scrutinizing for any anomalies, inaccuracies, or inconsistencies that may have emerged during the data preparation process.
- Transactions identified as fraudulent undergo cancellation; therefore, in the context of fraud detection, the utilization of the following columns—namely, oldbalanceOrg, newbalanceOrig, oldbalanceDest, and newbalanceDest—is deemed inappropriate see Fig.1. To present how dataset looks there we can see only 30 data, but overall analysis has been done approximately for 150 data.



```
= Table.RemoveColumns(#"Changed Type",{"step", "oldbalanceOrg", "newbalanceOrig", "oldbalanceDest", "newbalanceDest"})
```

	type	amount	nameOrig	nameDest	isFraud	isFlaggedFraud
1	PAYMENT	9839.64	C1231006815	M1979787155	0	0
2	PAYMENT	1864.28	C1666544295	M2044282225	0	0
3	TRANSFER	181.0	C1305486145	C553264065	1	0
4	CASH_OUT	181.0	C840083671	C38997010	1	0
5	PAYMENT	11668.14	C2048537720	M1230701703	0	0
6	PAYMENT	7817.71	C90045638	M573487274	0	0
7	PAYMENT	7107.77	C154988899	M408069119	0	0
8	PAYMENT	7861.64	C1912850431	M633326333	0	0
9	PAYMENT	4024.36	C1265012928	M1176932104	0	0
10	DEBIT	5337.77	C712410124	C195600860	0	0
11	DEBIT	9644.94	C1900366749	C997608398	0	0
12	PAYMENT	3099.97	C249177573	M2096539129	0	0
13	PAYMENT	2560.74	C1648232591	M972865270	0	0
14	PAYMENT	11633.76	C1716932897	M801569151	0	0
15	PAYMENT	4098.78	C1026483832	M1635378213	0	0
16	CASH_OUT	229133.94	C905080434	C476402209	0	0
17	PAYMENT	1563.82	C761750706	M1731217984	0	0
18	PAYMENT	1157.86	C1237762639	M1877062907	0	0
19	PAYMENT	671.64	C2033524545	M473053293	0	0
20	TRANSFER	215310.3	C1670993182	C1100439041	0	0
21	PAYMENT	1373.43	C20804602	M1344519051	0	0
22	DEBIT	9302.79	C1566511282	C1973538135	0	0
23	DEBIT	1065.41	C1959239586	C515132998	0	0
24	PAYMENT	3876.41	C504336483	M1404932042	0	0
25	TRANSFER	311685.89	C1984094095	C932583850	0	0
26	PAYMENT	6061.13	C1043358826	M1558079303	0	0
27	PAYMENT	9478.39	C1671590089	M58488213	0	0
28	PAYMENT	8009.09	C1053967012	M295304806	0	0
29	PAYMENT	8901.99	C1632497828	M33419717	0	0
30	PAYMENT	9920.52	C764826684	M1940055334	0	0

Fig. 1. Imported data.

In the initial analysis, a stacked column chart (Fig. 2) was created, depicting "nameOrig" on the X-axis, "count of amount" on the column X-axis, and "sum of is fraud" on the Y-axis. The chart highlights a green line, representing the money laundering threshold. Instances above the line are potential money laundering cases, providing a visual aid to identify areas of interest within the dataset.

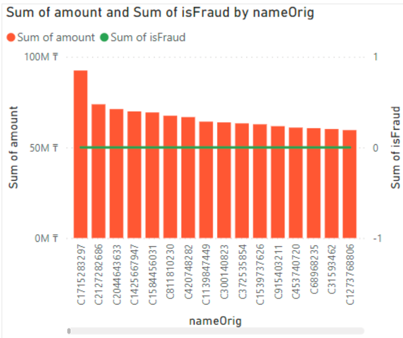


Fig. 2. Stacked column chart.

In the follow-up analysis, a line chart (Fig. 3) was generated with "nameOrig" values on the X-axis and "sum of isFlaggedFraud" on the Y-axis. The chart visually illustrates the cumulative sum of flagged fraud instances, offering insights into the dynamics of money laundering attempts associated with specific entities labeled as "nameOrig."

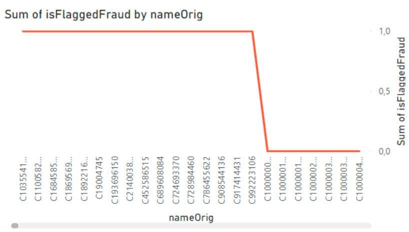


Fig. 3. Line chart.

For the next analysis, we created a pie chart. Here, an analysis based on transaction types and corresponding money laundering amounts was conducted. The results revealed that transfers constitute 49.88%, while withdrawals account for approximately 50.12% of the total amount see Fig.4. These two types emerge as the most prevalent and widely utilized methods for money laundering operations. The pie chart visually represents the proportion of each transaction type, facilitating an easy assessment of their contribution to the overall money laundering volume.

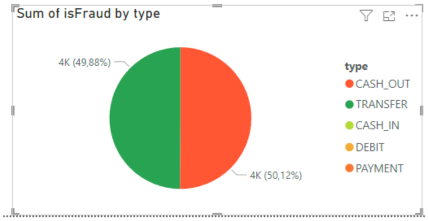


Fig. 4. Pie chart.

And the final stage of our analysis will be thoroughly examined to overview the total number of conducted transactions. This approach allows for a more detailed exploration of the diversity of financial operations and highlights their relative frequency in the context of the investigated time. Such analysis contributes to a better understanding of the overall dynamics of operations and identifies the most actively used types of transactions see Fig.5.

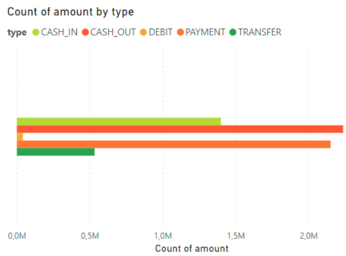


Fig. 5. Chart

The subsequent analyses, illustrated through stacked column charts, line charts, and pie charts, provided insights into potential money laundering activities. These visualizations highlighted patterns, thresholds, and trends, aiding in the identification of suspicious transactions.

Overall, the integration of Microsoft Power BI and meticulous data analysis techniques proved invaluable in uncovering patterns indicative of fraudulent or suspicious activities within the financial dataset.

4.2. *Application of ML in Money Laundering Detection*

Machine learning (ML) applications in combating the financing of terrorism are becoming increasingly important as financial crimes, including terrorism financing, become more sophisticated. Here are some ways in which machine learning is applied in the context of finance and terrorism: anomaly detection and network analysis. ML algorithms can analyse large volumes of financial transactions to identify unusual patterns or behaviours that may indicate terrorist financing. This includes detecting irregular transaction amounts, frequencies, and connections between entities. ML can be used to analyse complex networks of financial transactions, helping to identify hidden relationships and connections between individuals, organizations, and financial entities involved in terrorist financing [22].

It's important to note that while machine learning is a powerful tool, its effectiveness in combating terrorism financing is maximized when combined with human expertise, domain knowledge, and other traditional methods of investigation. Additionally, privacy and ethical considerations should be carefully addressed in the deployment of ML solutions for such sensitive applications [23, 24].

4.3. Integrating Microsoft Power BI with Python

In the evolving landscape of data analytics, the integration of Microsoft Power BI with Python has emerged as a powerful synergy, combining the robust visualization capabilities of Power BI with the versatility and analytical prowess of Python programming. This integration empowers data professionals to harness the strengths of both tools, unlocking new dimensions in data analysis, machine learning, and advanced visualization [25, 26].

Before delving into the integration process, ensure that Python is installed on your system. Subsequently, download and install Power BI Desktop and the Python Integration for Power BI to seamlessly marry the functionalities of these two robust platforms.

The integration of Power BI with Python extends beyond mere data transformations. Analysts and data scientists can leverage Python for predictive analytics, sentiment analysis, and other advanced use cases, seamlessly integrating these insights into Power BI reports and dashboards.

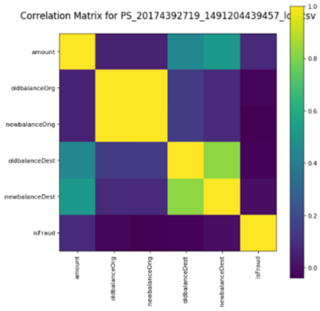


Fig. 6. Correlation matrix on Python

In conclusion, the marriage of Microsoft Power BI with Python fortifies the analytical capabilities of both tools, providing a holistic solution for professionals seeking to harness the power of data through advanced analytics and visualization.

5. Discussion

The article provides a detailed account of the data analysis process conducted using Microsoft Power BI to uncover patterns indicative of fraudulent or suspicious activities within a financial dataset. The discussion will delve into the key findings, the significance of each analysis, and the overall contribution of the integration of Power BI and meticulous data analysis techniques. The integration of Microsoft Power BI and meticulous data analysis techniques proved invaluable in uncovering patterns indicative of fraudulent or suspicious activities within the financial dataset. The visualizations provided clear insights into potential money laundering activities, facilitating informed decision-making and risk mitigation strategies. The systematic approach to data preparation, combined with the power of visual analytics, enhances the overall effectiveness of fraud detection efforts.

6. Conclusion

In conclusion, this article has embarked on a comprehensive exploration of money laundering detection, emphasizing the pivotal role of advanced financial monitoring and the challenges inherent in detecting increasingly sophisticated fraud techniques. The historical context of money laundering provides a backdrop for understanding its evolution and the subsequent need for robust detection methods.

In conclusion, this article serves as a valuable resource for researchers, analysts, and organizations grappling with the challenges of money laundering detection. By showcasing the integration of advanced tools and methodologies, it paves the way for future advancements in the ever-evolving landscape of financial fraud prevention. The presented analyses and visualizations stand as a testament to the power of technology-driven approaches in safeguarding the integrity of financial systems.

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